

RobinReal



Data Destroyers

Pelin - Thijs - Shirley - Umut

Hybrid Listing Search & Ranking



Search Strategy

Build a real-estate search system mapping natural queries to high-quality results.



Hard Criteria

Identify hard criteria, such as price, rooms, and location, to filter listings.



Validation

Retrieve and validate; only results satisfying all strict constraints are returned.



Soft Ranking

Rank valid listings by interpreting soft preferences (e.g., "bright," "modern").



Explainability

Provide a short explanation detailing why top results rank highly.

Data Normalization & Enrichment

OpenStreet Map

Geospatial, public transport info



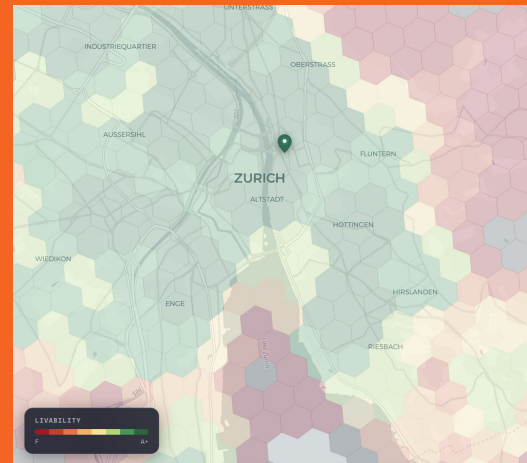
Crime Statistics

In Zürich



Strado.info

Livability Index



Quotes for illustration purposes only

Multimodal Bonus - Image Analysis



Process attributes for each listing based on the images uploaded

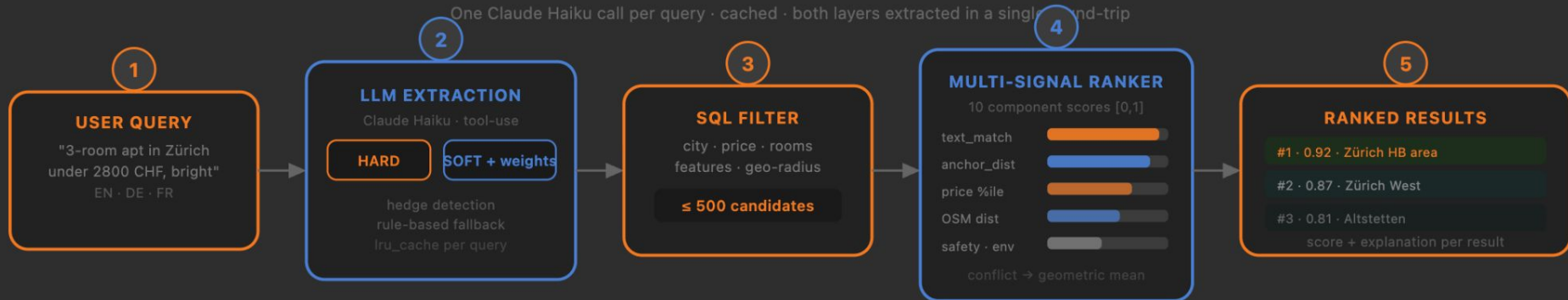
- View type (mountain, greenery)
- Brightness
- Size of rooms
- Modernity
- Summary of rooms visible

Automated extraction of subjective visual qualities.

Our Pipeline at a Glance

Query → Ranking Pipeline

One Claude Haiku call per query · cached · both layers extracted in a single end-trip



Query Understanding

- Extracts both **hard filters** and **soft preferences**.
- Primary path: structured LLM extraction with Claude
- Fallback path: rule-based parser.

Real Queries Mix Two Very Different Signals

EXAMPLE QUERY

"3-room apartment in Zürich under 2800 CHF , ideally with balcony , close to public transport "

HARD CONSTRAINTS

Must be satisfied — SQL WHERE clause

city · max price · min rooms
date · geo-radius · RENT / SALE

Empty results if violated → filter strictly



SOFT PREFERENCES

Influence ranking — never filter out

"bright" · "quiet" · "modern kitchen"
"close to ETH" · "good views"

Hedge words → ranking weight, not filter

Soft Filtering

- Conservative filtering before ranking.
- Removes obvious mismatches, such as parking-only objects when the user wants an apartment.
- Optionally trims expensive listings when the user explicitly asks for the cheapest option.
- Avoids over-filtering so good candidates are not lost too early.

Multi-Signal Ranking

Multi-Signal Ranker — 10 Components

Each signal scored [0,1] · LLM assigns per-query importance weights · min-max normalised to [0.05, 1.0]

text_match

Synonym hits of soft signals in listing title + description

w 3.0

environment

Parks, lakes, rivers nearby

w 1.4

feature_match

Structured feature flags (balcony, elevator, parking ...)

w 2.5

safety

Severity-weighted crime index per 1000 residents by city

w 1.2

anchor_distance

Haversine to POIs (ETH, HB Zürich ...) with commute-time decay

w 2.5

object_category

Listing type match (Studio, Loft, Maisonette ...)

w 1.0

price (context-relative)

Percentile within city + rooms cohort — CHF 2000 is cheap in Zürich, expensive in Delémont

w 1.5

image_analysis (Claude Vision + OpenCLIP)

Brightness · modernity · natural light · renovation quality · view type

w 0.4

structured_distance

OSM precomputed: transit stops · shops · schools · nightlife · noise buffers

w 1.5

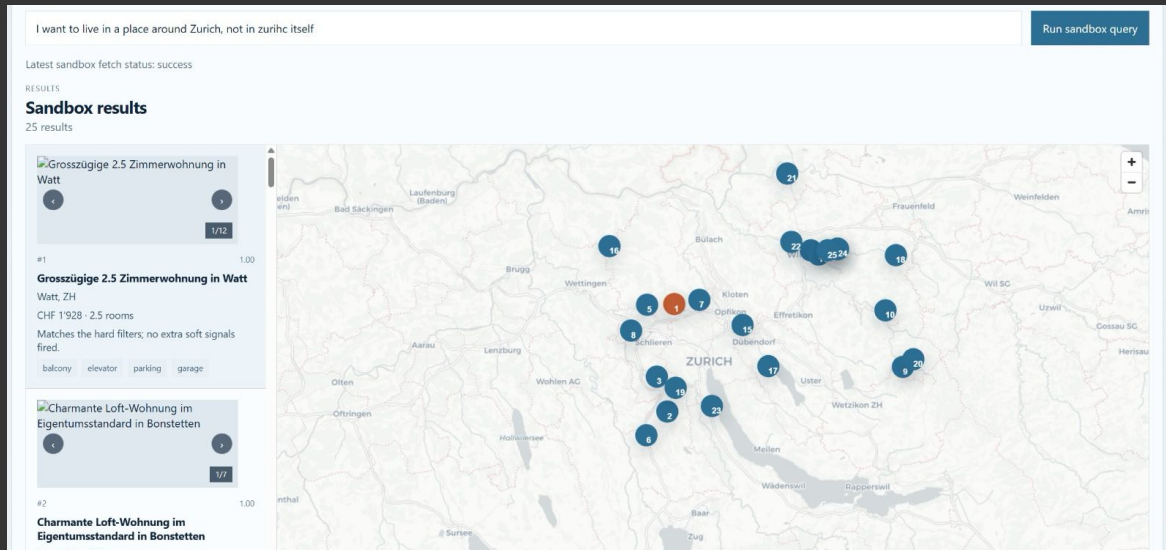
data_completeness

Fraction of key fields filled — rewards rich listings

w 0.3

Conflict detected (e.g. "cheap BUT central") → geometric mean replaces arithmetic mean · penalises one-sided wins

Special Cases



Implicit Preferences

Cues pointing to preferred features without explicit statement

"Affordable" → differs in context

Conflicts

Conflicting preferences handled by choosing the "midpoint"

Central vs cheap

Negations

"Near Zurich, not in Zürich"



#1

1.00

[Tauschwohnung] Tausche schöne Altbauwohnun in Wollishofen

Zürich

CHF 3'034 · 4.5 rooms

cheap; balcony pref; balcony; below the median for
Zürich

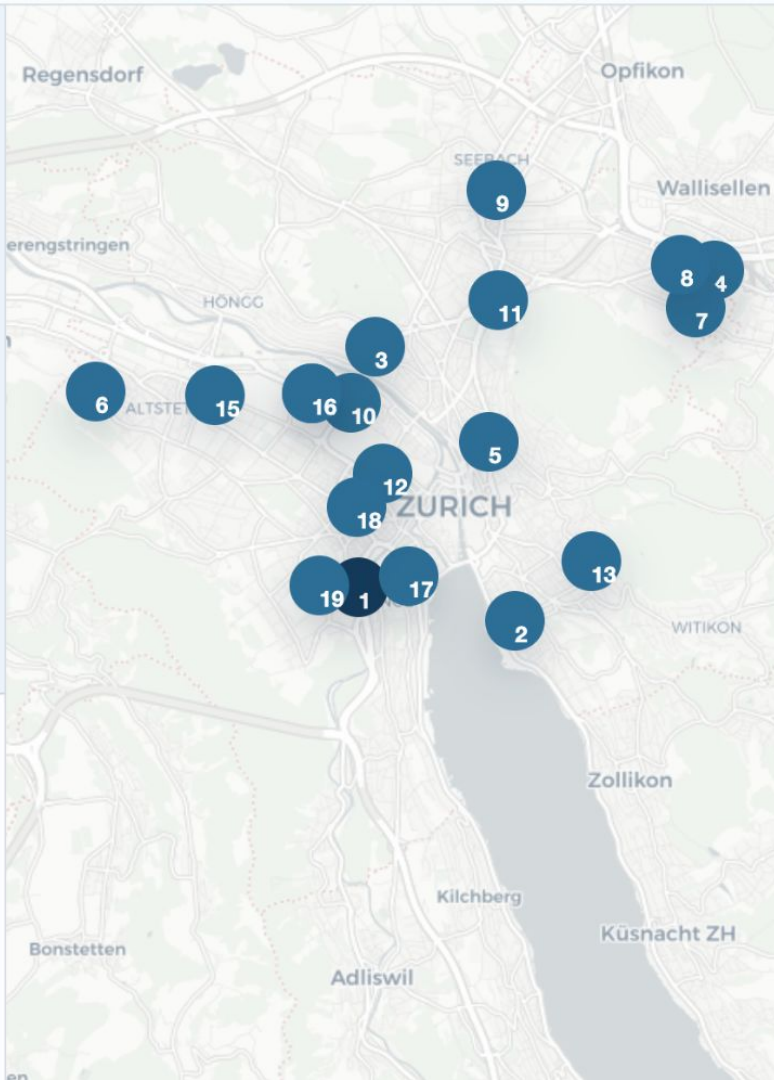
balcony



#2

0.92

Befristete 4-Zimmer-Wohnung in Zürich bis



Final Takeaway

- Fully deployed Fa + React widget + bridge
- Hard-filter precis the primary metr hedge detection i main contribution
- System is transpa every result inclu score breakdown evidence strings

Possible UI interface

AI-powered search for Switzerland

Describe your next home

Tell us what you're looking for in plain language. We'll find the best matches.

bright 3 room apartment |

Search →

Try searching for:

3-room flat in Zurich under 2800 CHF with balcony

Family apartment near good schools in Bern

Modern studio in Geneva, close to public transport

Quiet apartment with lake view in Lausanne

BASED ON YOUR PREFERENCES

Take a look at...



Zürich

TOP-Preis für eine temporäre Wohnlösung – befristet, all-inclusive...

CHF 3'990/mo



Dietlikon

Helle & moderne Familienwohnung!

CHF 2'790/mo



Kloten

Diese Wohnung sucht Sie!

CHF 1'855/mo

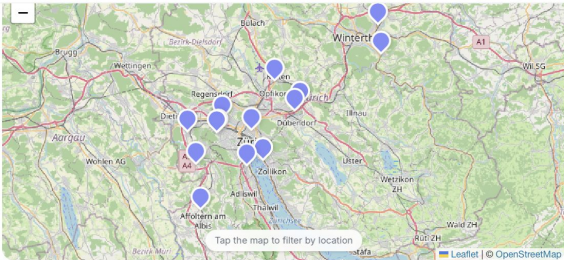
See all

Robin

New search

Filters

bright 3 room apartment*
14 listings in Switzerland



For Rent



Zürich, ZH

TOP-Preis für eine temporäre Wohnlösung – befristet, all-inclusive...

3 rooms • 70 m² • 03.02.2026

CHF 3'990 / mo

▲ 53.5% vs avg

For Rent



Dietlikon, ZH

Helle & moderne Familienwohnung!

4.5 rooms • 89 m² • 15.04.2026

CHF 2'790 / mo

▼ 20.3% vs avg

For Rent



Kloten, ZH

Diese Wohnung sucht Sie!

CHF 1'855 / mo

For Rent



Wetzikon, ZH

Diese Wohnung sucht Sie!

CHF 1'855 / mo



For Rent

Diese Wohnung sucht Sie!

Obstgartenstrasse 20, 8302 Kloten, ZH

CHF 1'855 / month ▼ 36% vs. area average

Average for this room type: CHF 2'900/mo

3.5 Rooms

63 m² Area

15.04.2026 Available

City Kloten

Canton ZH

Elevation 408 m a.s.l.

Nearest transit Zürich Wiedikon

About this listing

- Moderne Küche, - GK und GS - Modernes Bad - Wohnzimmer Parkett - Balkon - nahe Zentrum - Nahe öffentlichen Verbindungen - Nahe Einkaufsmöglichkeiten Die Fotos zeigen den Standard der Liegenschaft und können von der gezeigten Wohnung abweichen. Konnten wir Ihr Interesse wecken? Dann freuen wir uns auf Ihre Kontaktaufnahme!

Features

Balcony

Elevator

Parking

Garage

Fireplace

Child-friendly

Pets-allowed

Wheelchair-accessible

Minergie-certified

New-build

Neighbourhood (within 500 m)

Number of amenities within a 500 m radius

8 Transit stops

22 Shops

4 Parks

2 Schools

1

16

5